

Use-Case Flow Specification

Vaccination Cohort & Dose Scheduling System — Problem Statement #17

Use Case: UC-02 – Book Vaccination Slot (Dose 2)

Actor: Citizen Registrant

Includes: UC-03 – Validate Dose Interval

Extended by: UC-04 – Reschedule Slot (optional)

Preconditions

- The citizen is registered in the system and has an active account.
- The citizen has completed and been recorded for Dose 1.
- The Dose 1 administration date is on record in the system.

Postconditions

- A Dose 2 slot at a vaccination center is reserved and confirmed for the citizen.
- The selected center's slot capacity is decremented by 1.
- A booking confirmation is sent to the citizen.

Main Success Scenario

1. Citizen selects “Book Dose 2” from their dashboard.
2. System retrieves the citizen's Dose 1 date and calculates the elapsed interval in days.
3. System validates that the interval is at least 28 days (includes UC-03: Validate Dose Interval).
4. System displays available vaccination centers and open slots to the citizen.
5. Citizen selects a center and a preferred time slot.
6. System reserves the slot and decrements the center's remaining capacity by 1.
7. System sends a booking confirmation to the citizen (in-app and/or email/SMS).
8. Use case ends successfully.

Alternate Flow – 3a. Interval Not Met

- 3a1. System displays an “Not yet eligible” message showing the earliest valid booking date.
- 3a2. The booking/slot-selection screen is not shown; the citizen is redirected to their dashboard.
- 3a3. Citizen may optionally set a reminder notification for the eligible date.
- 3a4. Use case ends without a booking being created.

Alternate Flow – 6a. Slot No Longer Available

- 6a1. If the selected slot is taken by another citizen before reservation completes, the system displays a “Slot unavailable” message.
- 6a2. System returns the citizen to the updated list of available slots (Step 4).